Arista Access Node BAS-ACC-78 Optical Transport Failure SOP

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9946

Description

Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Arista Access Node BAS-ACC-78 at Basavanagudi POP is a critical failure domain that occurs due to the degradation of the optical signal strength over time or physical damage to the fiber optic cabling. This failure domain is unique to the Access role within the Arista equipment due to the reliance on the Optical Transport Network (OTN) protocol for high-speed transmission of data between nodes. The OTN protocol, specified in ITU-T G.709, uses SFP/XFP transceivers to transmit and receive optical signals over the fiber optic cables. The degradation of the SFP/XFP transceivers or damage to the fiber optic cables can lead to a reduction in the received optical power, resulting in a loss of signal integrity and ultimately, a failure of the optical transport. The Alarm Code ALARM_9946 is triggered when the received optical power falls below a certain threshold, indicating a major failure within the optical transport domain.

Pre-requisite Checks

- Verify that the Basavanagudi POP is operational and accessible

- Check the status of the redundant link to ensure that it is operational and can take over in case of a failure

- Notify the NOC team of the failure and request their assistance in dispatching remote hands to the Basavanagudi POP

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Optic Cable Damage	Medium
Rx/Tx Power Loss due to Optical Signal Attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces gigabitEthernet 1/1/1 detail | include 'rx bytes', Expected Output: rx bytes: 0, Description: This command is used to verify the received bytes on the gigabitEthernet interface. A zero value indicates a complete loss of signal, indicating a failure within the optical transport domain.

Command: show interfaces gigabitEthernet 1/1/1 detail | include 'tx bytes', Expected Output: tx bytes: 0, Description: This command is used to verify the transmitted bytes on the gigabitEthernet interface. A zero value indicates a complete loss of signal, indicating a failure
```

```
within the optical transport domain.

Command: show optics gigabitEthernet 1/1/1, Expected Output: Optics
Information for GigabitEthernet1/1/1, Description: This command is
used to verify the optical signal strength and quality on the
gigabitEthernet interface. A degraded signal quality or strength
indicates a failure within the optical transport domain.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Transceiver, Command: replace optics gigabitEthernet 1/1/1 with transceiver slot 1/1/1, Description: This step is used to replace the SFP/XFP transceiver on the gigabitEthernet interface. This is a permanent fix for the failure and requires a hardware replacement.

Step 2: Step: Verify the Optical Signal Strength and Quality, Command: show optics gigabitEthernet 1/1/1, Description: This step is used to verify the optical signal strength and quality on the gigabitEthernet interface after replacing the SFP/XFP transceiver. A stable and high-quality signal indicates a successful fix.

Step 3: Step: Verify the Network Connectivity, Command: ping 10.10.10.10, Description: This step is used to verify the network connectivity after replacing the SFP/XFP transceiver. A successful ping indicates a restored connection to the network.

Post-Check Verification

```
Command: show interfaces gigabitEthernet 1/1/1 detail | include 'rx
bytes', Expected Output: rx bytes: [non-zero value], Description: This
command is used to verify the received bytes on the gigabitEthernet
interface. A non-zero value indicates a restored signal, indicating a
successful fix.

Command: show interfaces gigabitEthernet 1/1/1 detail | include 'tx
bytes', Expected Output: tx bytes: [non-zero value], Description: This
command is used to verify the transmitted bytes on the gigabitEthernet
interface. A non-zero value indicates a restored signal, indicating a
successful fix.

Command: show optics gigabitEthernet 1/1/1, Expected Output: Optics
Information for GigabitEthernet1/1/1, Description: This command is
used to verify the optical signal strength and quality on the
gigabitEthernet interface. A stable and high-quality signal indicates
a successful fix.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Transceiver, Command: replace optics gigabitEthernet 1/1/1 with transceiver slot 1/1/1, Description: This step is used to restore the original SFP/XFP transceiver on the gigabitEthernet interface. This is a temporary fix to roll back to the original state before the replacement.

- Step: Verify the Original State, Command: show optics gigabitEthernet 1/1/1, Description: This step is used to verify the original state of the optical signal strength and quality on the gigabitEthernet interface. A stable and high-quality signal indicates a successful rollback.